

Lecture 5: Singular Value Decomposition (SVD)

INF1004 Linear Algebra (Part II)

Singapore Institute of Technology

1 Introduction and Motivation

In previous lectures, we studied **Eigendecomposition** (or Diagonalization). This process allows us to factorize a matrix A into the form:

$$A = PDP^{-1}$$

where D is a diagonal matrix of eigenvalues and P is the matrix of eigenvectors. However, this only applies to **square matrices**.

The Problem: A non-square matrix does not have a determinant, meaning eigenvalues and eigenvectors do not exist in the traditional sense.

The Solution: We use **Singular Value Decomposition (SVD)**. Every matrix $A \in \mathbb{R}^{m \times n}$ has an SVD of the form:

$$A = U\Sigma V^T$$

2 The Core Strategy: Choosing the Path

The products $A^T A$ and AA^T are always square and symmetric. The choice of which to diagonalize depends on the dimensions of A .

2.1 Case 1: $n < m$ (More rows than columns)

When A is "tall," $A^T A$ is smaller ($n \times n$) than AA^T ($m \times m$).

1. **Compute $A^T A$:** This $n \times n$ matrix is used to find V .
2. **Find Eigenvalues (λ_i):** Solve $|A^T A - \lambda I| = 0$.
3. **Construct Σ :** The singular values are $\sigma_i = \sqrt{\lambda_i}$. Place them in an $m \times n$ matrix.
4. **Find V :** Find and normalize the eigenvectors of $A^T A$. These are the columns of V .
5. **Find U :** Calculate the first n columns using $u_i = \frac{1}{\sigma_i} A v_i$.
6. **Pad U :** If $m > n$, find additional orthonormal vectors to complete the $m \times m$ matrix U .

2.2 Case 2: $m < n$ (More columns than rows)

When A is "wide," AA^T is smaller ($m \times m$) than $A^T A$ ($n \times n$).

1. **Compute AA^T :** This $m \times m$ matrix is used to find U .
2. **Find Eigenvalues (λ_i):** Solve $|AA^T - \lambda I| = 0$. Note that AA^T and $A^T A$ share the same non-zero eigenvalues.

3. **Construct Σ :** The singular values are $\sigma_i = \sqrt{\lambda_i}$. Place them in an $m \times n$ matrix.
4. **Find U :** Find and normalize the eigenvectors of AA^T . These are the columns of U .
5. **Find V :** Calculate the first m columns using $v_i = \frac{1}{\sigma_i} A^T u_i$.
6. **Pad V :** If $n > m$, find additional orthonormal vectors to complete the $n \times n$ matrix V .

3 Detailed Worked Example ($n < m$)

Evaluate the SVD of $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix}$. Since $n = 2$ and $m = 3$, we use $A^T A$.

3.1 Step 1: Eigenvalues and Σ

$$A^T A = \begin{bmatrix} 9 & 8 \\ 8 & 9 \end{bmatrix} \implies |A^T A - \lambda I| = (9 - \lambda)^2 - 64 = 0$$

Roots: $\lambda_1 = 17, \lambda_2 = 1$. Singular values: $\sigma_1 = \sqrt{17}, \sigma_2 = 1$.

$$\Sigma = \begin{bmatrix} \sqrt{17} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

3.2 Step 2: Right Singular Vectors (V)

$$\text{For } \lambda_1 = 17 : \begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

$$\text{For } \lambda_2 = 1 : \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies v_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}.$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

3.3 Step 3: Left Singular Vectors (U)

$$u_1 = \frac{1}{\sqrt{17}} A v_1 = \frac{1}{\sqrt{34}} \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix} \text{ and } u_2 = \frac{1}{1} A v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

To find u_3 , we need a vector orthogonal to u_1 and u_2 . Using the cross product:

$$u_3 = \text{normalize}(u_1 \times u_2) = \frac{1}{\sqrt{17}} \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}$$

4 Application: Recommender Systems

In a rating matrix A where rows are users and columns are items:

- U represents **User-to-Concept** similarity.
- Σ represents the **strength** of each concept.

- V^T represents **Item-to-Concept** similarity.

SVD allows us to predict missing ratings by identifying underlying patterns (latent factors) in how users rate specific categories of items.